# Statistical Inference and Modeling

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# Lectures

- Lecture 1: Introduction
- Lecture 2: Model selection
- Lecture 3: Resampling
- Lecture 4: Linear regression
- Lecture 5: Robust regression
- Lecture 6: Classification
- Lecture 7: Logistic regression
- Lecture 8: Neural networks
- Lecture 9: K-nearest neighbor
- Lecture 10: Bayesian decision, LDA, QDA, and naive Bayes
- Lecture 11: Support vector machine
- Lecture 12: Tree-based methods
- Lecture 13: Bagging, random forecast, boosting
- Lecture 14: Unsupervised learning and clustering algorithms
- Lecture 15: Unsupervised learning and PCA
- Lecture 16: Gaussian mixture model clustering
- Lecture 17: DBSCAN clustering

# 5. Robust Regression

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1 Table of Contents

2 Outliers in training data
- Outliers in measurements (y)
- Leverage points in input features

3 Outlier diagnostics
- Recap: linear regression
- Hat matrix (unusual $x$ values)
- Residual analysis
- Cook's distance

4 Robust regression
- Weighted least-squares
- Loss Functions for Robust Regression
- Regression M-estimators
- Median absolute deviation
- Iterative re-weighted least square

5 Summary

6 Appendix

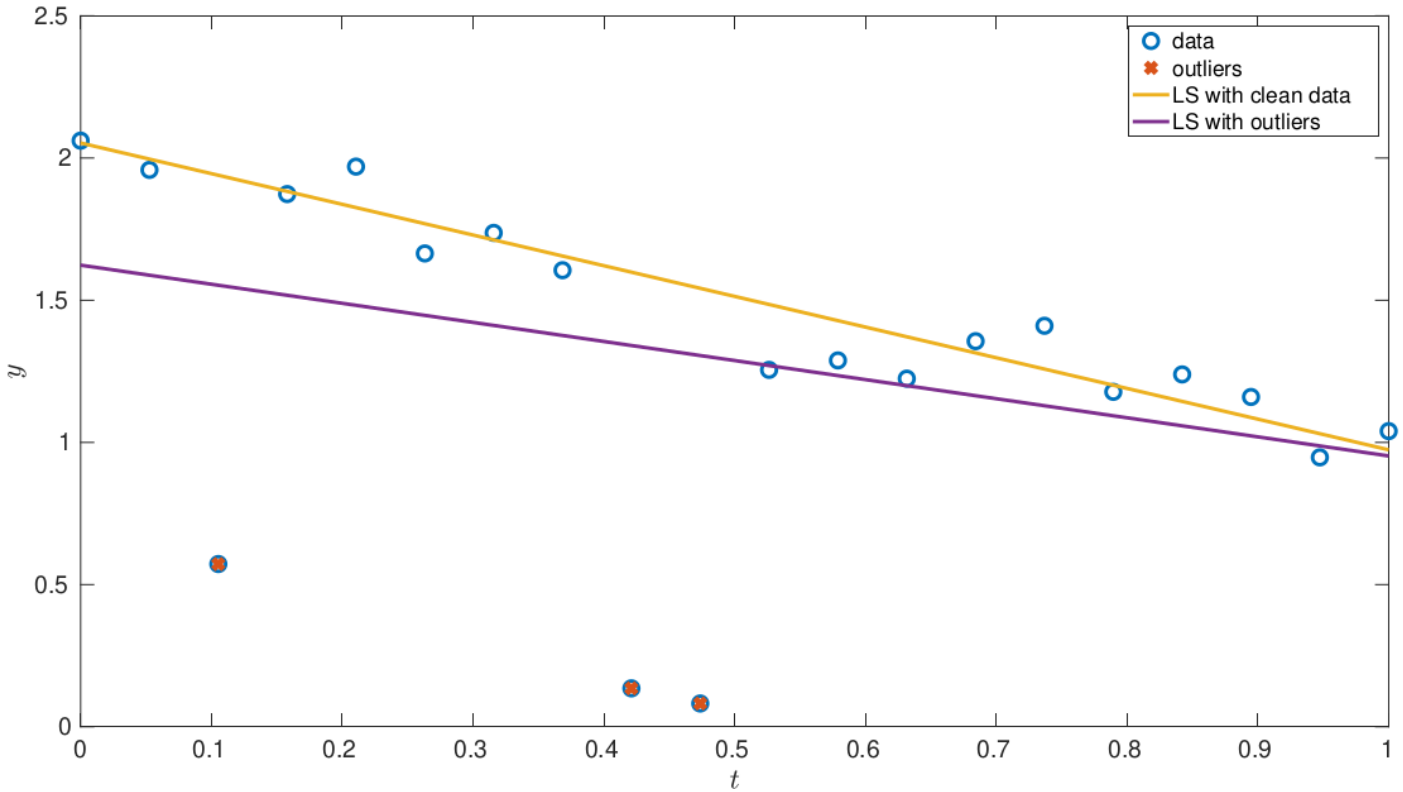
7 Index

# Outliers in training data

2 Outliers in training data
Outliers in measurements (y)
Leverage points in input features

# Outliers in training data

**Problem:** The output from linear regression can be impacted by outliers in training data [4, 5].



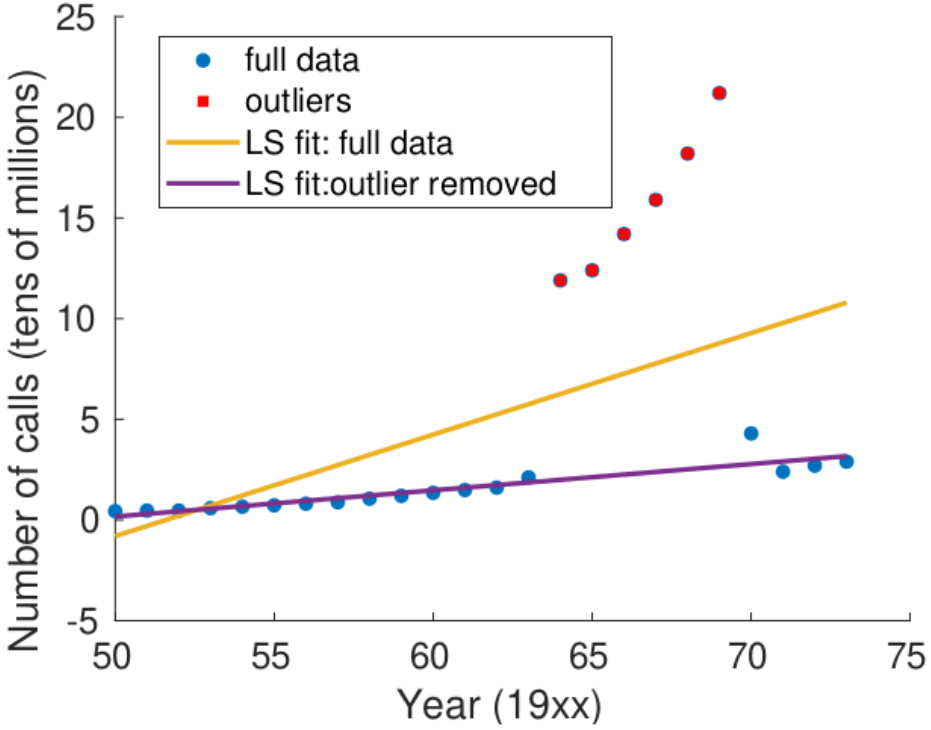

- Outliers: data points with extrememly diverging **measurements**.
- High leverages: data points with different mapping between intput and output prediction. Usually, the problem lies in **the input features**.

Robust regression can overcome the problems.

# Outliers in measurements (y)

**Example:** This is a problem of regression for the number of international phone calls from Belgium during 1950-1975 [4].

- Settings:
  - **Model**: $\#\text{calls} = f(\text{year})$
  - **Input** features: year
  - **Output** prediction: the number of calls (#calls)
- Nevertheless, the phone call contains outliers during 1964-1969.
- The slope is affected by the vertical outliers (tilted upward) .

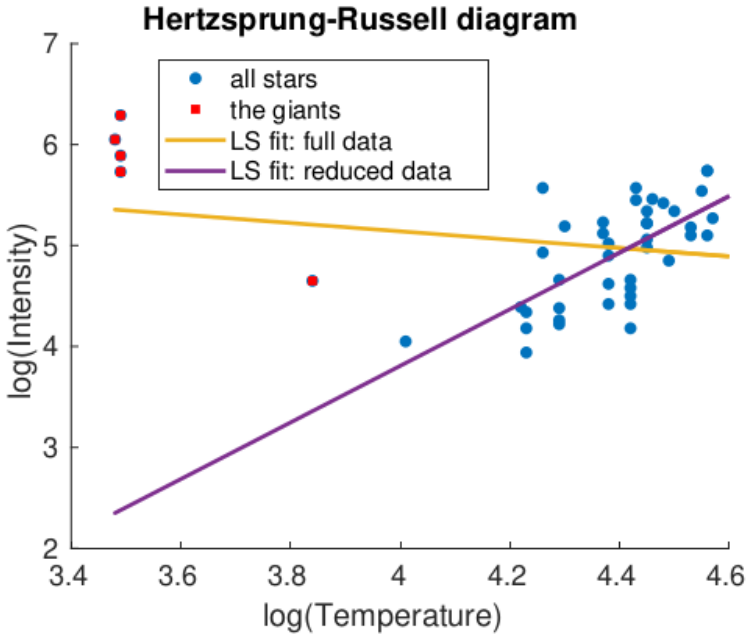



The number of international phone calls from Belgium during 1950-1975 [4]

# Leverage points in input features

**Example:** This is a problem of regression for the star intensity [5]

- Settings:
  - **Model**: intensity $= f(\text{temp})$
  - **Input** features: light intensity (intensity)
  - **Output** prediction: temperature (temp)
- Note that there are 5 giant stars. These stars have significantly lower temperature, which gives results in an inconclusive function mapping.
- Data points such as these are called *bad leverage points*. They are not error, but they contains other input features (such as size of the stars) that can impact the intensity differently from the temperature.




Hertzsprung-Russell diagram of the star cluster CYG OB1 (47 stars) [5]

# Outlier diagnostics

3 Outlier diagnostics
- Recap: linear regression
- Hat matrix (unusual $x$ values)
- Residual analysis
- Cook's distance

# How to remove these outliers

**Goal:** We can use the following methods [4] to identify potentially unusual outliers and remove them... We call these methods **outlier diagnostics**.

**Outlier diagnostics:** the followings are the methods for checking and removing the outliers in training data:

- Hat matrix
- Residual analysis
- Cook's distance

Generally, these will compute a regression based on deleting **one data point** and measure **the impact**. The impact is measured as a residual. If the residual is too large, they will be identified as outliers.

# Recap: linear regression

**Settings:** we have $m$ measurements, *i.e.*, $\boldsymbol{y} \in \mathcal{R}^m$ and $\boldsymbol{X} \in \mathcal{R}^{m \times n}$ where

$$\boldsymbol{X} = \begin{bmatrix} (\boldsymbol{x}^1)^T \\ (\boldsymbol{x}^2)^T \\ \vdots \\ (\boldsymbol{x}^m)^T \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_n^1 \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \vdots & \vdots \\ x_1^m & x_2^m & \dots & x_n^m \end{bmatrix}$$

$\boldsymbol{x}^j \in \mathcal{R}^n$ for $n$ being the number of feature dimension.
**Formulation:**

$$\boldsymbol{y} \approx \hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{\beta}$$

where $\boldsymbol{\beta}$ can be estimated by solving the least-square error minimization:

$$\boldsymbol{\beta}_{ls} = \underset{\boldsymbol{\beta}}{\textbf{argmax}} \; ||\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}||_2^2 = \sum_j^m (y_j - \sum_i^n x_i^j \beta_i)^2 \tag{1}$$

- $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}$ is called *the residual error*.
- $\boldsymbol{\beta}$ that give the smallest $||\boldsymbol{r}||_2$ is called *the least-squares solution*.
- $\boldsymbol{\beta}_{ls}$ can be obtained from finding $\boldsymbol{\beta}$ that gives the gradient $\nabla_\beta ||\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}||_2^2 = 0$.

$$\nabla_\beta ||\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}_{ls}||_2^2 = 2\boldsymbol{X}^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}_{ls}) = 0$$
$$\boldsymbol{\beta}_{ls} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y} \dots$$

If $\boldsymbol{X} \in \mathcal{R}^{m \times n}$ with $m \geq n$ is full rank, then *the least-squares solution* $\boldsymbol{\beta}_{ls}$ is unique [2]. Up until this point, both $\boldsymbol{X}$ and $\boldsymbol{y}$ are of the training data.

# Hat matrix (unusual $x$ values)

Given $\boldsymbol{\beta}_{ls}$, one can obtain the prediction $\hat{\boldsymbol{y}}^{\text{Test}}$ by multiplying $\boldsymbol{\beta}_{ls}$ with the input feature $\boldsymbol{X}^{\text{Test}}$. Note that we use superscript 'Test' to separate the training data components that is used to derive $\boldsymbol{\beta}_{ls}$.

$$\hat{\boldsymbol{y}}^{\text{Test}} = \boldsymbol{X}^{\text{Test}}\boldsymbol{\beta}_{ls} = \boldsymbol{X}^{\text{Test}}\boldsymbol{\beta}_{ls} = \boldsymbol{X}^{\text{Test}}\underbrace{(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}}_{\text{Train}}$$

The hat matrix of a linear regression is defined as

$$\boldsymbol{H} = \boldsymbol{X}^{\text{Test}}\underbrace{(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T}_{\text{Train}}$$

that maps the measurements $\boldsymbol{y}$ (from training data).
The matrix form can be written as follows:

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_{m'} \end{bmatrix}^{\text{Test}} = \boldsymbol{X}^{\text{Test}}\underbrace{(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}}_{\text{Train}} = \begin{bmatrix} h_{1,1} & h_{1,2} & \dots & h_{1,m} \\ h_{2,1} & h_{2,2} & \dots & h_{2,m} \\ \vdots & \vdots & \dots & \vdots \\ h_{m',1} & h_{m',2} & \dots & h_{m',m} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

- $h_{j,i}$ transfers the impact from the $i$th observation to $\hat{y}_j$.
- If we take derivative $\partial\hat{y}_j/\partial y_i$, then $h_{j,i}$ gives the effect of $y_i$ to $\hat{y}_j$.
- Because $\boldsymbol{H}$ is a matrix that is divided by $\boldsymbol{X}^T\boldsymbol{X} \in \mathcal{R}^{n\times n}$, it is possible that the diagonal coefficient $0 \le h_{j,j} \le 1$.

# Example

One can calculate the averaged value of diagonal coefficients in matrix $\boldsymbol{H}$ as follows:

$$\bar{h} = \sum_j h_{j,j} / \text{min}(\{m', m\})$$

which should be bounded by $n/m'$, *i.e.*, $\bar{h} \leq n/m'$.
First, $\text{min}(\{m', m\}) = m'$ because it is common that the numbers of observation at testing phase will be less than the one's at training.
But what about the denominator $n$ ? Let's see $\boldsymbol{X} \in \mathcal{R}^{m \times n}$. One can show the followings [3]:

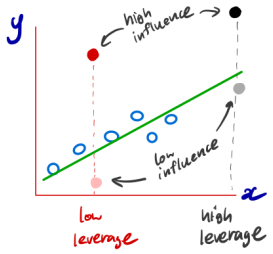
$$\sum_j h_{j,j} = \text{Tr}(\boldsymbol{H}) = \text{Tr}(\boldsymbol{X}^{\text{Test}} \underbrace{(\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T}_{\text{Train}}) \approx \text{Tr}(\boldsymbol{X}^T \boldsymbol{X} (\boldsymbol{X}^T \boldsymbol{X})^{-1}) = \text{Tr}(\boldsymbol{I}_{n \times n}) = n$$

Then, we define **high leverage data points** as any $h_{j,j}$ that are larger than the averaged value by some threshold:

*The $j^{th}$ data pair $(\boldsymbol{x}^j, y_j)$ is an high leverage, if $h_{j,j} > \frac{2n}{m'}$.*

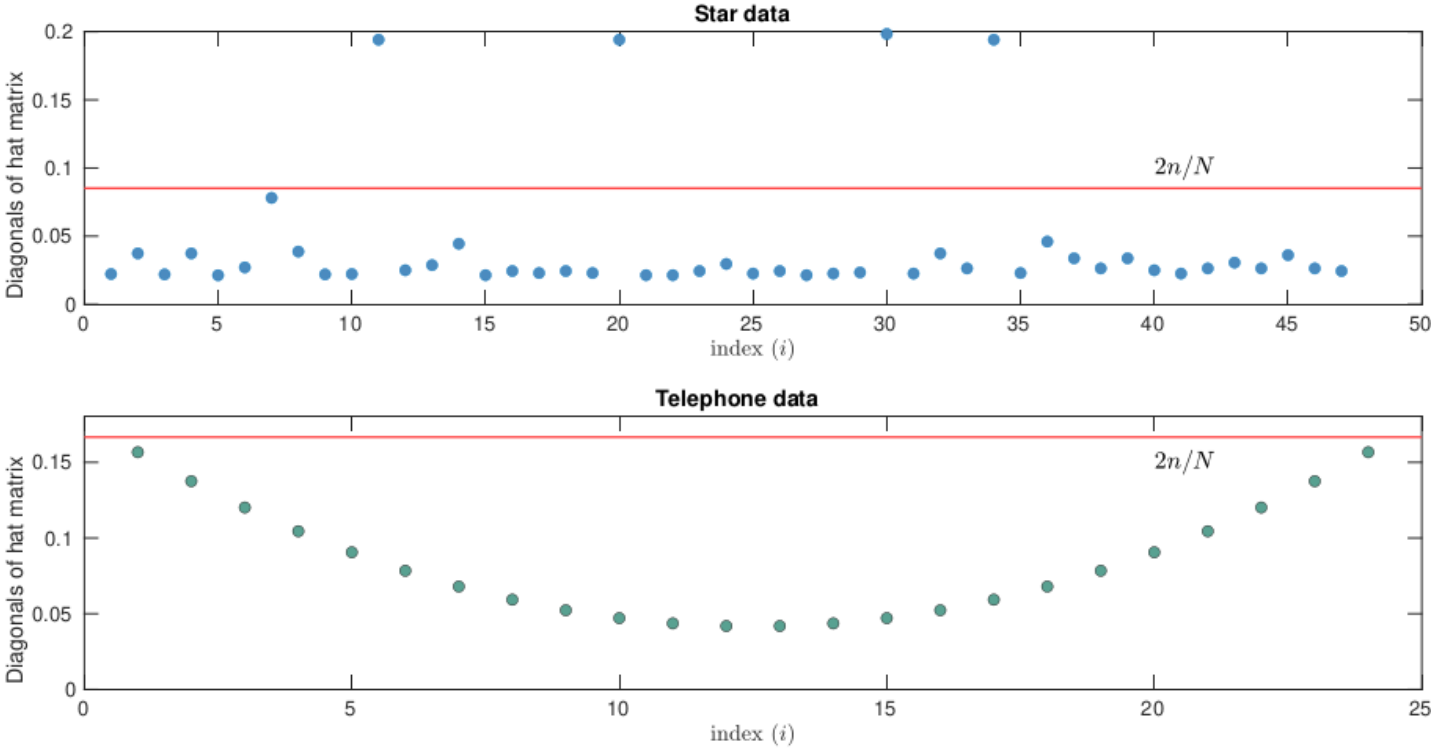
Nevertheless, the high leverage data points can have either **low** or **high** influence on the error:

- they have **low** influence on the error, if $y_j$ deviates from the norm by a small distance.
- they have **high** influence on the error, if $y_j$ greatly deviates from the norm.



# Example

The diagonal values of hat matrix (star and telephone datasets) $X$ is standardized. Red lines are the thresholding value at $2n/m'$...




- $h_{j,j}$'s of the giant stars are larger than $2n/m'$ ($2n/m'$=0.085).
- $h_{j,j}$'s of the telephone data are all under $2n/m'$, so no outliers are pointed out.
- Hat matrix is a function of $X$ can detect the outliers in **input features**.
- Whether or not the data point is influential depends on $y_i$.

# Idea of residual analysis

Let us consider $\boldsymbol{H}$ in the outlier analysis. In the ideal case, $\boldsymbol{X}^{\text{Test}}$ will be similar to the training's, that is, $\boldsymbol{X}^{\text{Test}} \approx \boldsymbol{X}$ in the hope that $\boldsymbol{H} \approx \boldsymbol{I}$

$$\boldsymbol{H} = \boldsymbol{X}^{\text{Test}}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T \approx \boldsymbol{I} \tag{2}$$

As a side note, one should be able to calculate a leverage $h_{jj}$ by $h_{j,j} = \underbrace{(\boldsymbol{x}^j)^T}_{\text{Test}}\underbrace{(\boldsymbol{X}^T\boldsymbol{X})^{-1}(\boldsymbol{x}^j)}_{\text{Train}}$.

- Eq.(2) appears similar to psuedo matrix inversion, which should guarantee that $\boldsymbol{H} = \boldsymbol{I}$. However, this may not be the case, especially if the input features are not a good representative for its predictive value $\hat{\boldsymbol{y}}$, or those testing input features are different from the training input features.
- Therefore, another way to diagnose for the outliers is to consider the residual from the identity matrix:

$$\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{y} \tag{3}$$

- Nevertheless, this approach lacks the consideration on the scale of $\boldsymbol{y}$. In the followings, we will explore two more concrete approaches to remove the outliers based on the residual vector $\boldsymbol{d}$.

# Residual analysis

An alternative methods for checking whether the $j^{\text{th}}$ data pair $(\boldsymbol{x}^j, y_j)$ is an outlier:

- Compute $\boldsymbol{\beta}_{ls}^{-j} \in \mathcal{R}^{N \times M-1}$ with the remaining data, *i.e.*,

$$\boldsymbol{\beta}_{ls}^{-j} = (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^{-j\,T} \boldsymbol{y}^{-j}$$

- Compute the fitted value after removing $y_j$ by $\hat{\boldsymbol{y}}^{-j} = \boldsymbol{X}\boldsymbol{\beta}_{ls}{}^{-j}$ where $\boldsymbol{X} \in \mathcal{R}^{m \times n}$ which is the same as the training data.
- Let's define $\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{\beta}_{ls}$. One can compute **the residual between the one with fully mapped data and the one with deleting pairs** as follows:

$$\boldsymbol{d}^{-j} = \boldsymbol{X}\boldsymbol{\beta}_{ls} - \boldsymbol{X}\boldsymbol{\beta}_{ls}{}^{-j} = \hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}^{-j}.$$

- For each coefficient $d_i^{-j}$ (where $\boldsymbol{d}^{-j} = [d_i^{-j}]_{i=1:m}$ and $d_i^{-j} := \hat{y}_i - \hat{y}_i^{-j}$), where $\hat{\boldsymbol{y}}^{-j} = [\hat{y}_i^{-j}]_{i=1:m}$, and $\hat{\boldsymbol{y}} = [\hat{y}_i]_{i=1:m}$. The standardized residual is defined as:

$$r_i^{-j} = \frac{d_i^{-j}}{\text{RSE}(d_i^{-j})}.$$

  where $\text{RSE}(d_i^{-j})$ is the residual standard error of $d_i^{-j}$: $\text{RSE}(d_i^{-j}) = \sqrt{\frac{\sum_{i=1}^{m}(\hat{y}_i - \hat{y}_i^{-j})^2}{m-n}}$. Here, we use $m-n$ for $n$ is the number of parameters in the linear regression.
- Potential outliers are the one with $r_i^{-j} > 3$ (threshold).

# Outlier diagnostics — Cook's distance

Settings...

- The estimated parameters after deleting $(\boldsymbol{x}^j, y_j)$ are ${\boldsymbol{\beta}_{ls}}^{-j} \in \mathcal{R}^{n\times 1}$
- The fully mapped target $\hat{\boldsymbol{y}} = \boldsymbol{X}^{\text{Test}}\boldsymbol{\beta}_{ls} = \boldsymbol{X}\boldsymbol{\beta}_{ls}$
- The mapped target after deleting $(\boldsymbol{x}^j, y_j)$: $\hat{\boldsymbol{y}}^{-j} = \boldsymbol{X}^{\text{Test}}{\boldsymbol{\beta}_{ls}}^{-j} = \boldsymbol{X}{\boldsymbol{\beta}_{ls}}^{-j}$
- The residual from the fully mapped target $\boldsymbol{d}^{-j} = \hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}^{-j} \in \mathcal{R}^{m\times 1}$

**Cook's distance** measuring how far $\boldsymbol{\beta}_{ls}$ changes when $(\boldsymbol{X}_i, y_i)$ is removed:

$$
\begin{aligned}
D_j &= \frac{(\hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}^{-j})^T(\hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}^{-j})}{n \cdot \text{MSE}_y(\hat{y})} \\
&= \frac{(\boldsymbol{X}\boldsymbol{\beta}_{ls} - \boldsymbol{X}\boldsymbol{\beta}_{ls}^{-j})^T(\boldsymbol{X}\boldsymbol{\beta}_{ls} - \boldsymbol{X}\boldsymbol{\beta}_{ls}^{-j})}{n \cdot \text{MSE}_y(\hat{y})} \\
&= \frac{(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})^T(\boldsymbol{X}^T\boldsymbol{X})(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})}{n \cdot \text{MSE}_y(\hat{y})}
\end{aligned}
$$

- Note that the denominator is changed to $n \cdot \text{MSE}_y(\hat{y})$ where $n$ is the rank of $\boldsymbol{X}$.
- $\text{MSE}_y(\hat{y})$ is the (empirical) mean square error of the regression model: $\text{MSE}_y(\hat{y}) = \frac{\boldsymbol{e}^T\boldsymbol{e}}{m-n}$, where $\boldsymbol{e}$ is the residual vector defined in 3.
- $D_j$ measures the effect of the $j$th observation on the fitted vector and coefficients. High $D_j$ indicates that the observation could be outlier (threshold can be $4/n$ or $3 \cdot \bar{D}$)

# Cook's distance

The cook's distance

$$D_j = \frac{(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})^T (\boldsymbol{X}^T \boldsymbol{X})(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})}{n \cdot \mathsf{MSE}_y(\hat{y})} \tag{4}$$

has an equivalent form, which is expressed using the leverage ($0 \leq h_{j,j} \leq 1$)

$$D_j = \frac{{e_j}^2}{n \cdot \mathsf{MSE}_y(\hat{y})} \frac{h_{j,j}}{(1 - h_{j,j})^2}. \tag{5}$$

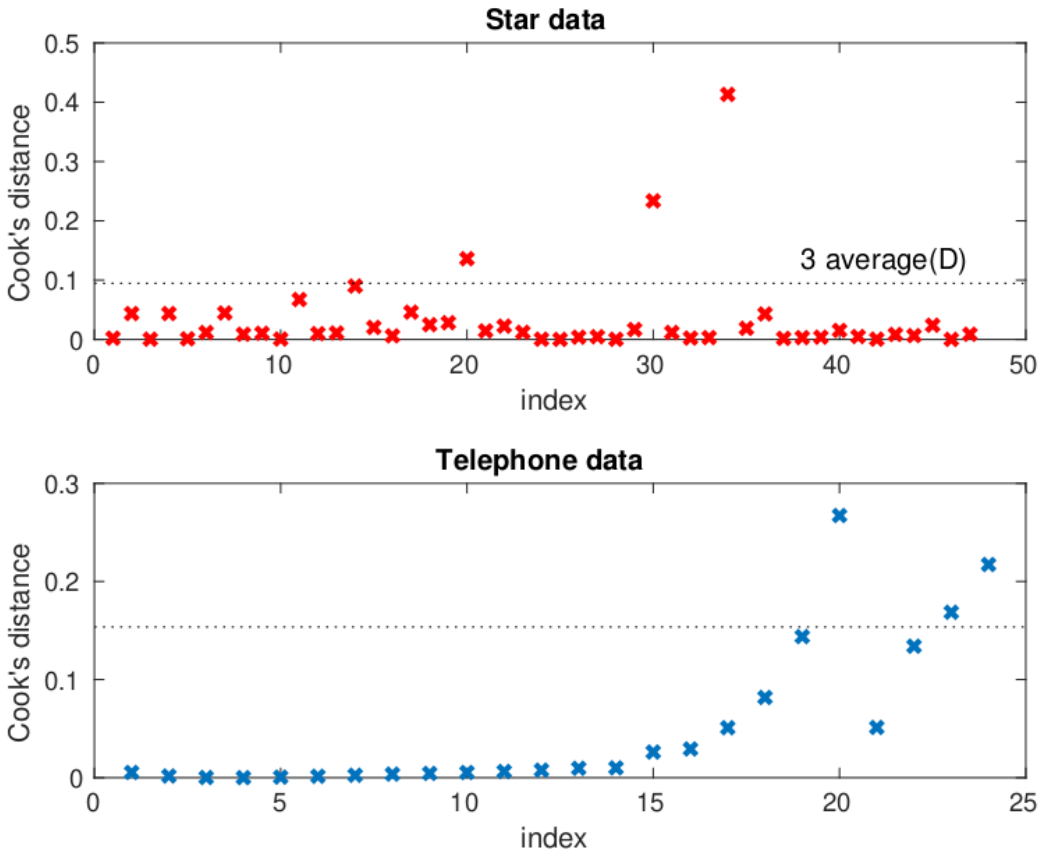
where $e_j$ is a coefficient in the residual vector $\boldsymbol{e} = [e_j]_{j=1:m} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{y}$. The proof that Eq.(4) is equivalent to Eq.(5) is provided in Appendix (P. 33).

Why you need to know this ...?

- Eq.(5) is more efficient to compute. You can verify this by flop counts.
- Because the impact of the bad leverages will reflect through $\hat{\boldsymbol{y}}$, but the measurement outliers will further extend the difference between $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$. Note that both will contribute to the high $D_j$ as shown in Eq.(4).
- However, Eq.(5) separates the impact of measurement outliers from the impact of the bad leverages (input feature outliers). This gives a better view towards identifying problems in the linear regression.

# Example of Cook's distance

Cook's distance calculated on star and telephone datasets:




- star indices of high $D_i$ are 20, 30, 34 (three of the giant stars).
- telephone indices of high $D_i$ are 20, 23, 24 (years 1969, 1972, 1973).

# Robust Regression

4 Robust regression
- Weighted least-squares
- Loss Functions for Robust Regression
- Regression M-estimators
- Median absolute deviation
- Iterative re-weighted least square

# Weighted least-squares

**Settings:** we have $m$ measurements, *i.e.*, $\boldsymbol{y} \in \mathcal{R}^m$ and $\boldsymbol{X} \in \mathcal{R}^{m \times n}$ where

$$\boldsymbol{X} = \begin{bmatrix} (\boldsymbol{x}^1)^T \\ (\boldsymbol{x}^2)^T \\ \vdots \\ (\boldsymbol{x}^m)^T \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_n^1 \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \vdots & \vdots \\ x_1^m & x_2^m & \dots & x_n^m \end{bmatrix}$$

$\boldsymbol{x}^j \in \mathcal{R}^n$ for $n$ being the number of feature dimension.

The weighted least-squares problem is defined as follows:
Let $\boldsymbol{W}$ be a positive definite matrix that can be factorized as $\boldsymbol{W} = \boldsymbol{L}^T\boldsymbol{L}$. A weighted least-square (WLS) problem is

$$\underset{\beta}{\textbf{argmin}}\ (\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{y})^T\boldsymbol{W}(\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{y}) = ||\boldsymbol{L}(\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{y})||^2$$

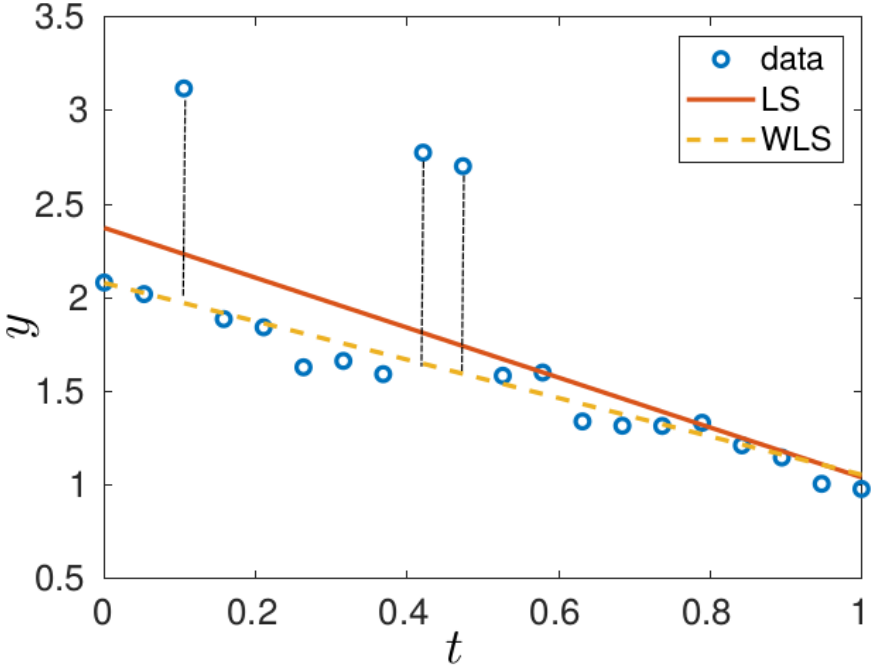
Similar to the simple linear regression, we can solve for $\boldsymbol{\beta}$ by applying gradient with respect to $\boldsymbol{\beta}$, and equate the equation to zero.

$$\begin{aligned} 2\boldsymbol{X}^T\boldsymbol{L}^T\boldsymbol{L}(\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{y}) &= 0 \\ \boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta} &= \boldsymbol{X}^T\boldsymbol{W}\boldsymbol{y} \\ \boldsymbol{\beta}_{wls} &= (\boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{W}\boldsymbol{y} \qquad \text{(if } \boldsymbol{X} \text{ is full rank).} \end{aligned}$$

# Interpretation of weighted least-squares

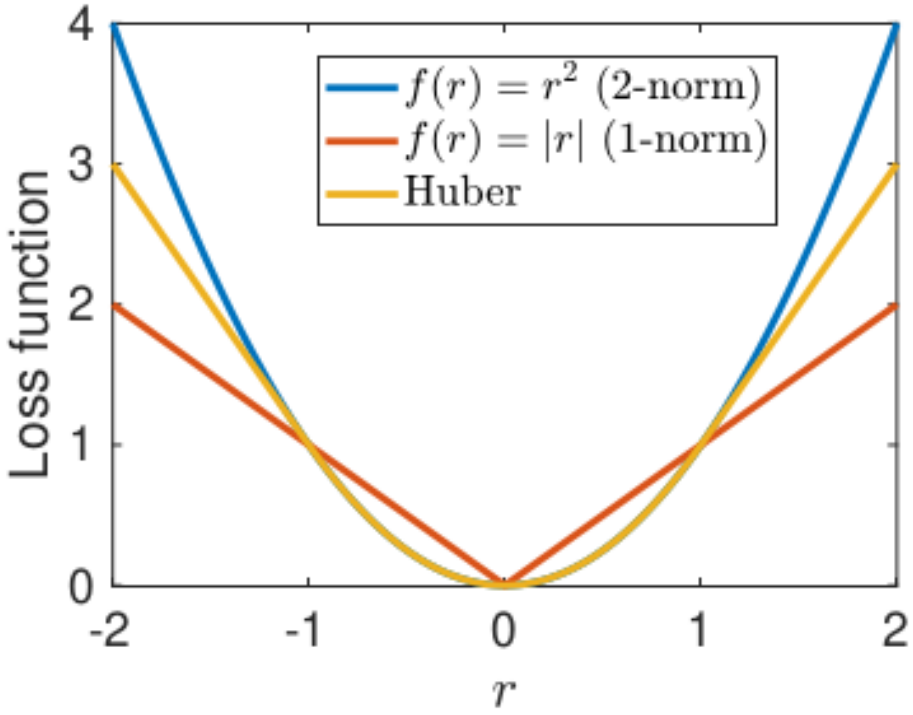
As samples 3, 9, 10 are outliers, we can suppress them by applying $\boldsymbol{W} = \text{diag}(w_1, w_2, ..., w_m) = \text{diag}(l_1^2, l_2^2, ..., l_m^2)$ onto the weighted least-square objective: $\sum_{j=1}^{m} w_j (y_j - \boldsymbol{x}^{j\,T}\beta)^2$.

- Use relatively low $w_3, w_9, w_{10}$ to penalize samples 3, 9, 10.
- The weights can adapt to the outliers which improves the robustness over the linear regression (denoted as the ordinary least square or LS).



# Loss functions for Robust Regression

- $l_2$-norm has the highest penalty on the large residual $|r| > 1$.
- However, the residual, if too large, will dominate the objective loss, which leads to the incorrect regression.
- Instead Huber and $l_1$ norms are often used in robust regression.
- These losses ( $l_2$, $l_1$, Huber ) can all be formulated as convex problem.




$$r = \boldsymbol{y} - \boldsymbol{X}\beta$$

$$\text{Huber}(r) = \begin{cases} (1/2)r^2, & |r| \leq M \\ M(|r| - M/2), & |r| > M \end{cases}$$

# Regression M-estimators

Generally, the regression model can be generalized into the M-estimator regression formulation defined as follows

$$\boldsymbol{\beta}_M = \underset{\boldsymbol{\beta}}{\textbf{argmin}} \sum_{j=1}^{m} \rho\left(\frac{r_j(\boldsymbol{\beta})}{\hat{\sigma}}\right) \tag{6}$$

where $\hat{\sigma}$ is a predefined scale.

- For $\rho(r) = r^2$, the M-estimator Eq.(6) becomes a least-square problem, and solving this equation will give $\boldsymbol{\beta}_M = \boldsymbol{\beta}_{ls}$.
- For $\rho(r) = |r|$, the M-estimator Eq.(6) becomes the least absolute deviation (LAD)

$$\beta_{lad} = \underset{\boldsymbol{\beta}}{\textbf{argmin}} \sum_{j=1}^{m} \frac{|y_j - \boldsymbol{x}^{j^T}\boldsymbol{\beta}|}{\hat{\sigma}}$$

  where solving the above equation will give $\boldsymbol{\beta}_M = \boldsymbol{\beta}_{lad}$. Here, $\hat{\sigma}$ is the predefined scale value.
- The predefined scale can be estimated using median absolute deviation:

$$\hat{\sigma} = k \cdot \text{MAD}_{y_j}(\hat{y}_j)$$

  where $k$ is the scale factor $\approx 1.4826$, and MAD is shorten for the median absolute deviation function $\text{MAD}_{y_j}(\hat{y}_j) := \text{Median}(|\hat{y}_j - y_j|)$ where $\hat{y}_j = \boldsymbol{x}^{j^T}\boldsymbol{\beta}_M$ and $y_j$ is the reference.

# Median absolute deviation

Let $\hat{x}^j$ be an estimate to the data point $x^j$ in $\{x^1, x^2, ..., x^m\}$. The median absolute deviation (MAD) function from its median is defined as:

$$\text{MAD}_{\tilde{x}}(x^j) = \text{Median}(|x^j - \tilde{x}|)$$

where $\tilde{x}$ is the median of the set $\{x^1, x^2, ..., x^m\}$.

**To use MAD as the estimator**, one have to multiply a scale factor $k$ to the median absolute deviation (MAD) function:

$$\hat{\sigma} = k \cdot \text{MAD}_{\tilde{x}}(x^j)$$

Some properties:

- Median absolute deviation (MAD) function is a robust estimation function for the scale parameter.
- $k$ is the scale factor, which depends on the data distribution. For a normally distributed data, $k$ is taken to be $k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$.
- Derivatoin: (next) how do we know that $k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$. ?

# Median absolute deviation (scale factor $k$)

To use MAD as the estimator for standard deviation ($\hat{\sigma}$), one have to multiply a scale factor $k$ to the median absolute deviation (MAD) function:

$$\hat{\sigma} = k \cdot \text{MAD}$$

For any normally distributed data, $k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$.

Below ... we show why ....

- Scale factor $k$ is $1/$Boundary, where the Boundary is the deviated value that cover $x$% of the data distribution.
- Let's defind the boundary $\pm$MAD that covers 50% of the standard normal distribution (between $1/4$ and $3/4$). Suppose that $X$ is any random variable.
- The normalized variable $Z = \frac{X-\mu}{\sigma}$

$$P(Z \leq \frac{\text{MAD}}{\sigma}) = P(\frac{|X' - \mu|}{\sigma} \leq \frac{\text{MAD}}{\sigma})$$
$$= \Phi\left(\frac{\text{MAD}}{\sigma}\right) - \Phi\left(-\frac{\text{MAD}}{\sigma}\right) = 2\Phi\left(\frac{\text{MAD}}{\sigma}\right) - 1 = \frac{1}{2}$$

  We use the fact that $\Phi(-x) = 1 - \Phi(x)$.
- Then, $\Phi(\frac{\text{MAD}}{\sigma}) = 3/4$, and $\frac{\text{MAD}}{\sigma} = \Phi^{-1}(3/4) = 0.67449$.
  The scale factor $k$ is

$$k = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826$$

# Iterative re-weighted least square

**Iterative re-weighted least square**: an iterative method that is used for solving the weighted least-square [1]:

$$\boldsymbol{\beta}^{(t+1)} = \underset{\boldsymbol{\beta}}{\textbf{argmin}} \sum_{j=1}^{m} w_j^{(t)} ({\boldsymbol{x}^j}^T \boldsymbol{\beta}^{(t)} - y_j)^2$$

where $\boldsymbol{\beta}^{(t+1)}$ is updated in each iteration. The weight $w_j^{(t)}$ is also updated in a way of mitigating the influence of outliers.

# Iterative re-weighted least square

$$\boldsymbol{\beta}^{(t+1)} = \underset{\boldsymbol{\beta}}{\textbf{argmin}} \sum_{j=1}^{m} w_j^{(t)} (\boldsymbol{x}^{j\,T} \boldsymbol{\beta}^{(t)} - y_j)^2$$

Generally, the iterative process consists of four steps:

1. Estimate $\boldsymbol{\beta}^t$ with a linear regression and compute the residual.
2. Find the high leverage coefficients and estimate the standard deviation of the residual.
3. Compute the weights to suppress the effects of these coefficients.
4. Recompute $\boldsymbol{\beta}^{(t+1)}$ with the new weights.

# Iterative re-weighted least square process (details) [5]

1 Start with calculating $\boldsymbol{\beta}^{(t)}$ and the residual vector

$$\boldsymbol{r} = [r_j]_{j=1:m}$$

where $r_j = \boldsymbol{x}^{j\,T}\boldsymbol{\beta}^{(t)} - y_j$.

2 Compute the leverage values $\boldsymbol{h} = [h_{j,j}]_{j=1:m} = \text{Diag}(\boldsymbol{H})$ where $\boldsymbol{H} = \boldsymbol{X}\boldsymbol{\beta}^t$.

3 Compute a normalized residual. There are two options to calculate the residual:

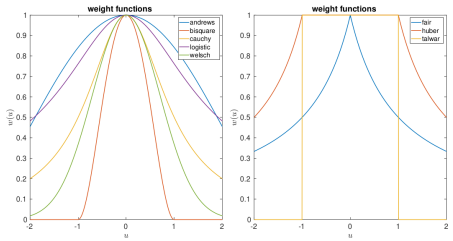
- $u_j = \frac{r_j}{\sqrt{1-h_{j,j}}}$
- $u_j = \frac{r_j}{K\hat{\sigma}\sqrt{1-h_{j,j}}}$

where $K$ is a tuning constant, and $\hat{\sigma} := k \cdot \text{MAD}_{y_j}(\hat{y}_j)$ and $k = 1.4826$.

4 Compute the robust weights $w_i$ as a function of $u_j$ (larger $u_j$, smaller $w_j$).
Bisquare is an example of the robust weight function. It puts zero weight on large residual.

$$\text{Bisquare:} \qquad w_j = \begin{cases} (1-u_j^2)^2, & |u_j| < 1 \\ 0, & |u_j| \geq 1 \end{cases}$$

$$\boldsymbol{W} = \text{diag}(w_1, ..., w_m)$$



# Iterative re-weighted least square

5 Update $\boldsymbol{\beta}^{(t+1)}$ by solving weighted least square estimation.

$$\underset{\beta}{\textbf{argmin}} \sum_{j=1}^{m} w_j^{(t)} (\boldsymbol{x}^{j\,T} \boldsymbol{\beta} - y_j)^2$$

Then, we can update $\boldsymbol{\beta}^{(t+1)}$ with $\boldsymbol{\beta}^{(t+1)} = (\boldsymbol{X}^T \boldsymbol{W} \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{W} \boldsymbol{y}$

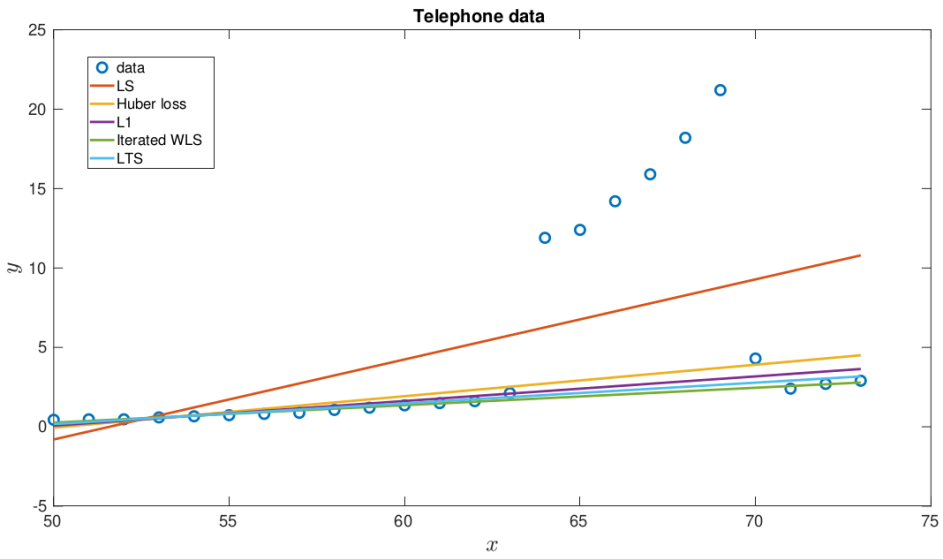
6 Update $\hat{y}_j$ with new $\boldsymbol{\beta}^{(t+1)}$, *i.e.*, $\hat{y}_j = \boldsymbol{x}^{j\,T} \boldsymbol{\beta}^{(t+1)}$ for $j = 1, ..., m$.

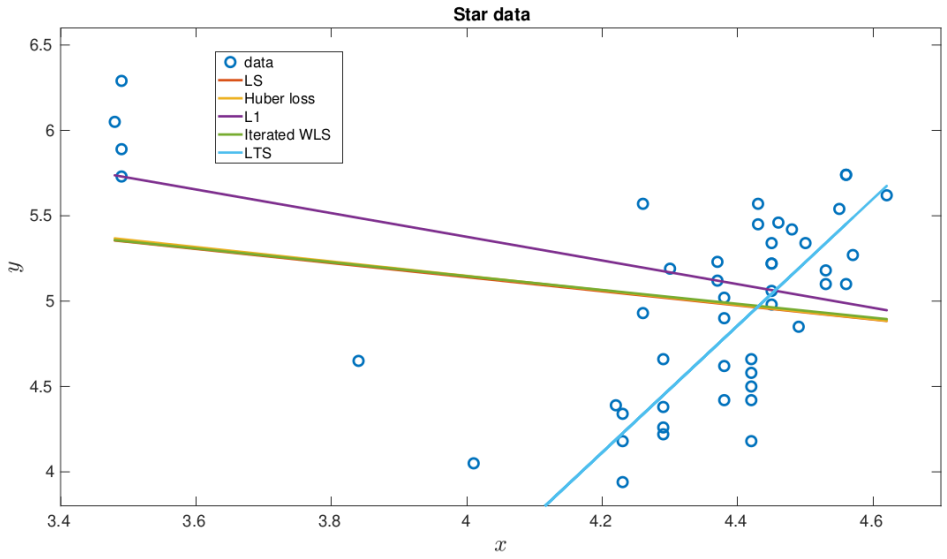
7 Calculate the error.

$$e = \sum_{j=1}^{m} (\hat{y}_j - y_j)^2.$$

8 Iteration stops if the fit converges, or the maximum number of the iterations is reached, go to the second step.

# Result using robust methods




- Telephone dataset has outliers in $y$-direction; robust methods can ameliorate the effects of outliers quite well
- Star dataset has bad leverage points; most robust methods (WLS, huber, L1) do not perform well on this dataset.
- LTS (least trimmed square estimator) finds a subset of observations with small residuals and estimate $\hat{\beta}$ based on that subset of data (more detail in P. J. Rousseeuw book)

# Summary

- We consider two important factors that can lower the performance of linear regression:
  1. measurement outliers
  2. bad leverage (input feature outliers)
- There are three main approaches to handle such outliers:
  1. outlier analysis (outlier removal methods): hat matrix, residual analysis, Cook's distance
  2. robust estiamtion methods: weighted least square, M-estimator.
  3. iterative method: iterative re-weighted least square
- **Outlier analysis (outlier removal methods)** checks for outliers and remove them.
- **Robust estiamtion methods** uses loss functions that put less emphasize on the outliers.
- **Iterative re-weighted least square** adjusts the weight function according to fitted residual in each step
- Other methods/algorithms exist: least median square (LMS), least trimmed squares estimator (LTS), random sample consensus (RANSAC)
- Python coding: https://github.com/GabbySuwichaya/Statistical-Learning-EE575/blob/master/Tutorial2/main.ipynb

# Appendix: expressing cook's distance using the leverage

The proof that Eq.(4) is equivalent to Eq.(5) is provided in Appendix, *i.e.*,

$$D_j = \frac{(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})^T(\boldsymbol{X}^T\boldsymbol{X})(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})}{n \cdot \mathsf{MSE}_y(\hat{y})} = \frac{e_j^2}{n \cdot \mathsf{MSE}_y(\hat{y})} \frac{h_{j,j}}{(1-h_{j,j})^2}$$

where $e_j$ is a coefficient in the residual vector $\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{y}$.
We can partition $\boldsymbol{X}^T\boldsymbol{X} \in \mathcal{R}^{n\times n}$ as follows:

$$\boldsymbol{X}^T\boldsymbol{X} = \sum_{i=1}^{m} \boldsymbol{x}^i \boldsymbol{x}^{i\,T} = \boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j} + \boldsymbol{x}^j \boldsymbol{x}^{j\,T}$$

Similarly, we can partition $\boldsymbol{X}^T\boldsymbol{y} \in \mathcal{R}^{n\times n}$ as follows:

$$\boldsymbol{X}^T\boldsymbol{y} = \sum_{i=1}^{m} \boldsymbol{x}^i y_i = \boldsymbol{X}^{-j\,T}\boldsymbol{y}^{-j} + \boldsymbol{x}^j y_j$$

Also ... $\boldsymbol{X} = \begin{bmatrix} (\boldsymbol{x}^1)^T \\ (\boldsymbol{x}^2)^T \\ \vdots \\ (\boldsymbol{x}^m)^T \end{bmatrix} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_n^1 \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \vdots & \vdots \\ x_1^m & x_2^m & \dots & x_n^m \end{bmatrix}$, $\boldsymbol{X}^{-j} = \begin{bmatrix} (\boldsymbol{x}^1)^T \\ (\boldsymbol{x}^2)^T \\ \vdots \\ (\boldsymbol{x}^{m-1})^T \end{bmatrix}$, $\boldsymbol{y}^{-j} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{m-1} \end{bmatrix}$

# Expressing cook's distance ... (cont')

Let's consider $(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})$

$$\begin{aligned}
\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j} &= \boldsymbol{\beta}_{ls} - \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{X}^{-j\,T}\boldsymbol{y}^{-j} \\
&= \boldsymbol{\beta}_{ls} - \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\left(\boldsymbol{X}^{T}\boldsymbol{y} - \boldsymbol{x}^{j}y^{j}\right) \\
&= \boldsymbol{\beta}_{ls} - \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{X}^{T}\boldsymbol{y} + \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{x}^{j}y^{j}
\end{aligned}$$

Compute $\left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}$ using Sherman–Morrison formula:

$$\begin{aligned}
\left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1} &= \left(\boldsymbol{X}^{T}\boldsymbol{X} - \boldsymbol{x}^{j}\boldsymbol{x}^{j\,T}\right)^{-1} \\
&= \left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1} + \frac{\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j}\boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}}{1 - \boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j}}
\end{aligned}$$

$$\begin{aligned}
\left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{x}^{j} &= \left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j} + \frac{\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j}\boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j}}{1 - \boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j}} \\
&= \left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j} + \frac{\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j}h_{j,j}}{1 - h_{j,j}} \\
&= \frac{\left(\boldsymbol{X}^{T}\boldsymbol{X}\right)^{-1}\boldsymbol{x}^{j}}{1 - h_{j,j}}
\end{aligned}$$

Let's go back ...

$$\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j} = \boldsymbol{\beta}_{ls} - \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y} + \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{x}^j y_j \tag{7}$$

Thus, we have $\left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{x}^j y_j = \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j y_j}{1-h_{j,j}}$.

Then, let's consider $\left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y}$

$$\begin{aligned}
\left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y} &= \left(\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1} + \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j\boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}}{1-\boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j}\right)\boldsymbol{X}^T\boldsymbol{y} \\
&= \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y} + \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j\boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}}{1-\boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j}\boldsymbol{X}^T\boldsymbol{y} \\
&= \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y} + \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j\boldsymbol{x}^{j\,T}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}}{1-h_{j,j}}\boldsymbol{X}^T\boldsymbol{y} \\
&= \boldsymbol{\beta}_{ls} + \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j\boldsymbol{x}^{j\,T}\boldsymbol{\beta}_{ls}}{1-h_{j,j}}
\end{aligned}$$

$$\begin{aligned}
\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j} &= \boldsymbol{\beta}_{ls} - \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y} + \left(\boldsymbol{X}^{-j\,T}\boldsymbol{X}^{-j}\right)^{-1}\boldsymbol{x}^j y_j \\
&= \boldsymbol{\beta}_{ls} - \left(\boldsymbol{\beta}_{ls} + \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j\boldsymbol{x}^{j\,T}\boldsymbol{\beta}_{ls}}{1-h_{j,j}}\right) + \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j y_j}{1-h_{j,j}} \\
&= -\frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j}{1-h_{j,j}}\left(\boldsymbol{x}^{j\,T}\boldsymbol{\beta}_{ls} - y_j\right)
\end{aligned}$$

Let's consider the nominator of $D_j$ ( Eq.(4))

$$
\begin{aligned}
\text{Nom}(D_j) &= (\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j})^T (\boldsymbol{X}^T\boldsymbol{X})(\boldsymbol{\beta}_{ls} - \boldsymbol{\beta}_{ls}^{-j}) \\
&= \left( \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j}{1-h_{j,j}} ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y_j) \right)^T (\boldsymbol{X}^T\boldsymbol{X}) \left( \frac{\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{x}^j}{1-h_{j,j}} ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y_j) \right) \\
&= \frac{1}{(1-h_{j,j})^2} ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y_j)^T {\boldsymbol{x}^j}^T \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1} (\boldsymbol{X}^T\boldsymbol{X}) \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1} \boldsymbol{x}^j ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y^j) \\
&= \frac{1}{(1-h_{j,j})^2} ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y_j)^T {\boldsymbol{x}^j}^T \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1} \boldsymbol{x}^j ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y_j) \\
&= \frac{h_{j,j}}{(1-h_{j,j})^2} ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y_j)^T ({\boldsymbol{x}^j}^T\boldsymbol{\beta}_{ls} - y_j) \\
&= \frac{h_{j,j}}{(1-h_{j,j})^2} ({e_j}^2)
\end{aligned}
$$

Recall that $\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{y} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}_{ls}$.

1 Table of Contents
2 Outliers in training data
Outliers in measurements (y)
Leverage points in input features
3 Outlier diagnostics
Recap: linear regression
Hat matrix (unusual $x$ values)
Residual analysis
Cook's distance
4 Robust regression
Weighted least-squares
Loss Functions for Robust Regression
Regression M-estimators
Median absolute deviation
Iterative re-weighted least square
5 Summary
6 Appendix
7 Index

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